

Linear Algebra I Quiz 1 Solution

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1. (a) The zero vector 0_V of V is the zero function $\mathbf{0}$, and $\mathbf{0}(1) = 0$, so $0_V = \mathbf{0} \in W$. Given $f, g \in W$, $f(1) = g(1) = 0$, and $(f + g)(1) = g(1) + g(1) = 0 + 0 = 0$. Thus, $f + g \in W$. Given $a \in F, h \in W$, $(ah)(1) = a \cdot h(1) = a \cdot 0 = 0$, so $ah \in W$. Therefore, W is a vector subspace of V .

- (b) If the equation

$$a(x - 1) + b(x^2 - 1) = 0$$

is satisfied for some $a, b \in F$, then $bx^2 + ax - (a + b) = 0$, which implies that $a = b = 0$. Thus, $\{x - 1, x^2 - 1\}$ are linear independent. It suffices to show that $\{x - 1, x^2 - 1\}$ spans W . For convenience, we put $f(x) = x - 1$, $g(x) = x^2 - 1$. Since $f(1) = g(1) = 0$, for any $a, b \in F$, we have

$$(af + bg)(1) = a \cdot f(1) + b \cdot g(1) = 0 + 0 = 0.$$

Therefore, $\text{span}\{f, g\} \subseteq W$. On the other hand, for any $h \in W$, $h(1) = 0$ and $\deg h \leq 2$, so there exists $q \in F[x]$ with $\deg q \leq 1$, such that $h = fq$ (by division algorithm). Furthermore, q can also be written as $(x + 1)r + s$ for some $r, s \in F$ (since $\deg q \leq 1$). Combining the result, h can be written as

$$h = fq = f((x + 1)r + s) = f \cdot (x + 1)r + fs = rg + sf.$$

Thus $h \in \text{span}\{f, g\}$. Consequently, $W \subseteq \text{span}\{f, g\}$.

2. (a) If the equation

$$a(1, 1, 0) + b(0, 1, 1) + c(1, 0, 0) = (0, 0, 0)$$

is satisfied for some $a, b, c \in F$, then (a, b, c) is a solution of the system of linear equations:

$$\begin{cases} a + c = 0 \\ a + b = 0 \\ c = 0 \end{cases}$$

Which forces that $a = b = c = 0$. Thus, $\{u_1, u_2, u_3\}$ is linear independent.

- (b) Given any $v = (x, y, z)$, we want to find $a, b, c \in F$ such that

$$a(1, 1, 0) + b(0, 1, 1) + c(1, 0, 0) = (x, y, z).$$

Equivalently, we want to find a solution of the system of linear equation:

$$\begin{cases} a + c = x \\ a + b = y \\ c = z \end{cases}$$

A solution is given by $a = x - z, b = y + z - x, c = z$.

3. (a) Since $S, T \subseteq S \cup T$, we have $\text{span}(S), \text{span}(T) \subseteq \text{span}(S \cup T)$. Note that $\text{span}(S), \text{span}(T)$ are vector subspaces of V , so by the minimality of sums, $\text{span}(S) + \text{span}(T) \subseteq \text{span}(S \cup T)$. Conversely, since $S \cup T \subseteq \text{span}(S) + \text{span}(T)$ and $\text{span}(S) + \text{span}(T)$ is a subspace, then by the minimality of spans, $\text{span}(S \cup T) \subseteq \text{span}(S) + \text{span}(T)$.
- (b) Since that $S \cap T \subseteq \text{span}(S) \cap \text{span}(T)$, and that $\text{span}(S) \cap \text{span}(T)$ is a vector subspace of V . By the minimality of spans,

$$\text{span}(S \cap T) \subseteq \text{span}(S) \cap \text{span}(T).$$

Let $V = \mathbb{R}^2$, $S = \{(0, 1), (1, 0)\}$, $T = \{(2, 0)\}$, then $\text{span}(S \cap T) = \text{span}(\emptyset) = \{0_V\}$ and $\text{span}(S) \cap \text{span}(T) = V \cap \text{span}(T) = \text{span}(T) \supsetneq \{0_V\}$.